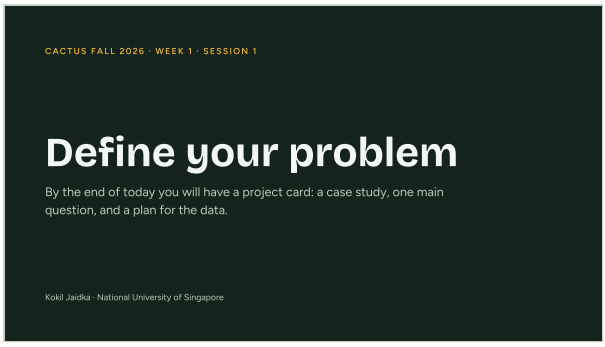
CACTUS Week 1: Define Your Problem



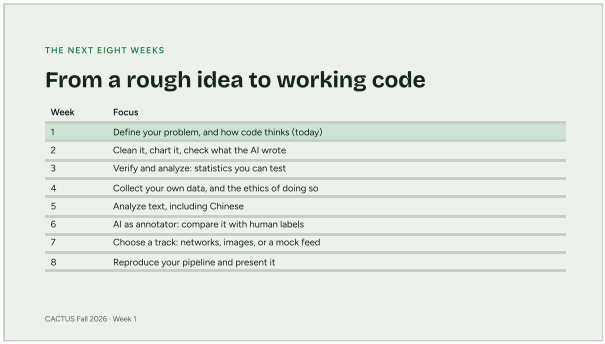
Slide 1

TIME 2 min (0:00)

SAY Welcome. This is an eight-week, hands-on Python series. No coding experience is needed, it is not for credit, and each week builds on the last. Today is about your project, not code.

DO Introduce yourself and the other facilitators by name. Post the course site link in the chat. Ask people to turn cameras on if they are comfortable.

WATCH FOR Late joiners. Do not restart for them; a facilitator catches them up in the chat.



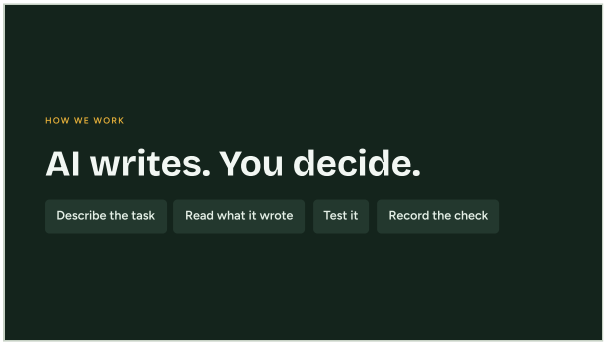
Slide 2

TIME 3 min

SAY Here is the whole arc, one line per week. Today you finish a project card. Everything later builds on it. Topics may shift slightly after the planning survey.

DO Read the table out in one pass. Point at row 1 (today). Do not explain later weeks in detail.

IF ASKED "Do I need to know Python?" No. Weeks 1 and 2 are written for absolute beginners.



Slide 3

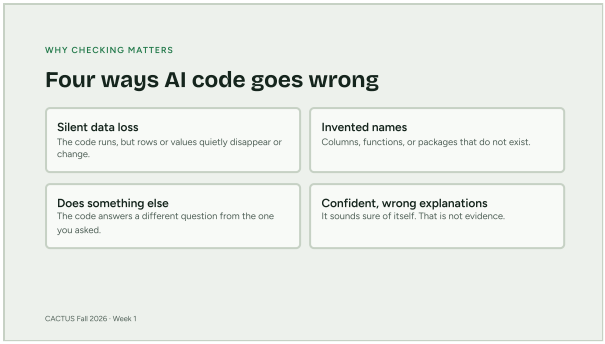
TIME 8 min (includes the live demo)

SAY You will use AI every week. The skill is not typing code from memory. It is describing a task precisely, reading what the AI wrote, testing it on a small sample, and recording how you checked.

DO (live demo, 3 min) Open any AI chat. Paste a tiny made-up table (two or three rows with columns id, country, score). Then ask: "Using my table, show the average of the column called 'sentiment_score'." The column does not exist. The AI will usually invent code anyway. Read its answer aloud and ask: what would you not be able to tell from this answer?

SAY A confident answer is not evidence. That is the whole course in one sentence.

WATCH FOR If the AI says the column does not exist, that is fine: say "good, this time it caught it" and move on. Have a second prompt ready (ask for a column that is nearly right, like 'Score').



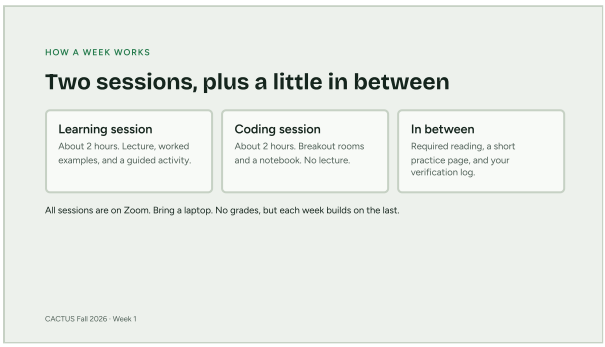
Slide 4

TIME 4 min

SAY Four ways AI code goes wrong. We practice catching each one. You do not need to memorize them.

DO Read the four cards. For each, give one sentence of example: silent loss (rows quietly disappear), invented names (a column that does not exist), does something else (answers a different question), confident wrong explanations.

IF ASKED "Should I not use AI then?" Use it freely. The course is about checking it, not avoiding it.



Slide 5

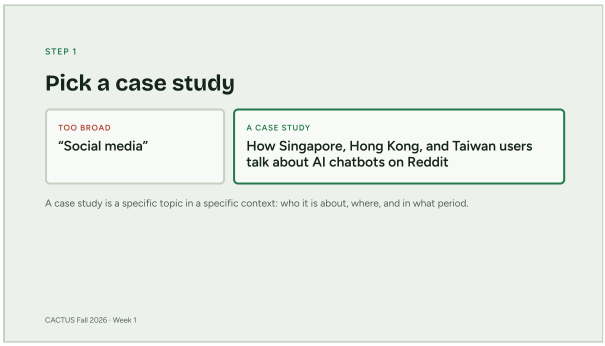
TIME 3 min

SAY Each week has a learning session, a coding session, and a little in between: required reading, a short practice page, and your verification log.

DO Name the facilitators again and say what they do: they visit breakout rooms and watch the chat.

IF ASKED about the reading The textbook sections are listed on the course site. Required, not graded.

WATCH FOR Anxiety about workload. Say: the weekly work is designed to take a few hours, and you can always ask.



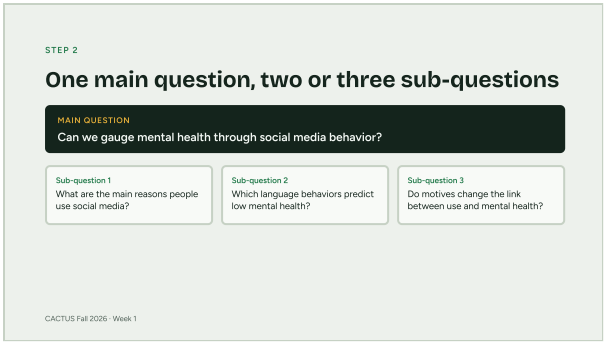
Slide 6

TIME 6 min (0:20)

SAY Step 1: a case study. A specific topic in a specific context. Not "social media" but a named topic, place, and platform.

DO Ask the room for one more example of a too-broad topic and help turn it into a case study live. Choose someone willing; do not cold-call.

WATCH FOR People who say they have no topic yet. Say: pick something you care about for now. You can change it. There is a fallback dataset for anyone with no data.



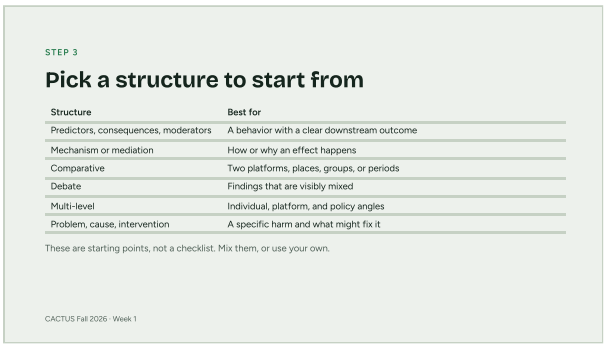
Slide 7

TIME 7 min

SAY Step 2: one main question and two or three sub-questions. The sub-questions break the big one into pieces you can answer.

DO Read the mental-health example. For each sub-question, ask: what would you count or compute to answer this? Get one answer from the room.

IF ASKED "How many sub-questions?" Two or three. More than that and the project will be too big for eight weeks.



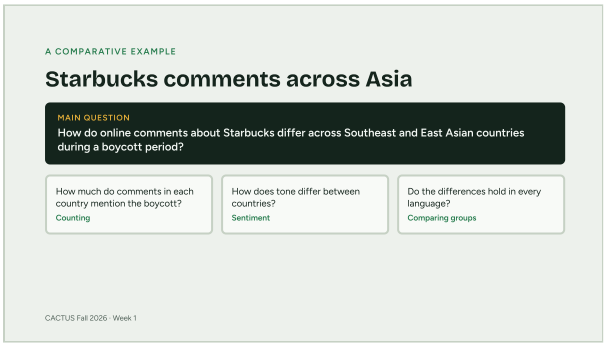
Slide 8

TIME 6 min

SAY Six structures to start from. These are starting points, not a checklist. Pick one that fits, mix, or use your own.

DO Do not read the whole table. Name the six, then pick one row (comparative) and say when you would use it. Tell people the project guide has a worked example for each.

IF ASKED "Which is best?" None. Pick the one that fits your case study.



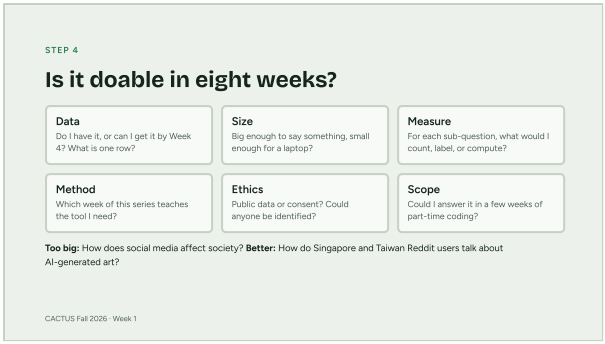
Slide 9

TIME 6 min (ends 0:45)

SAY Here is one worked example with the Starbucks comments you will see in Week 2. Each sub-question maps to something we will teach: counting, sentiment, comparing groups.

DO Read the main question and the three sub-questions. Point out the green label under each one: that is the skill. Ask: could you say what you would measure for each?

SAY That is the test of a good sub-question.



Slide 10

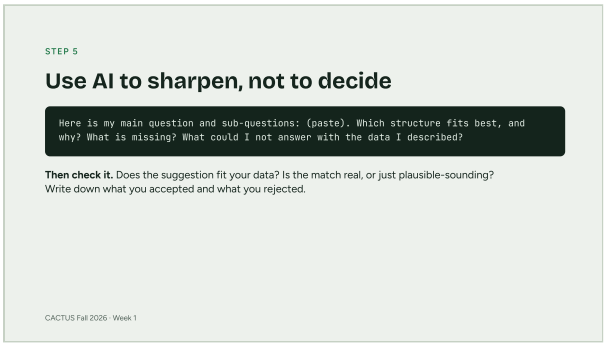
TIME 12 min (0:45 to 0:57)

SAY Step 4: is it doable in eight weeks? Six quick checks. If you answer no or not sure to any, that tells you what to work on, it is not a failure.

DO Go through the six cards. Spend longest on Data (what is one row?) and Measure (what would you count, label, or compute?). Read the too-big versus better example at the bottom. Pause for questions.

IF ASKED about ethics Week 4 covers it properly. For now: is the data public, and could anyone be identified?

WATCH FOR Ambitious projects. Say: it is fine to start big and narrow it. We will revise it together.



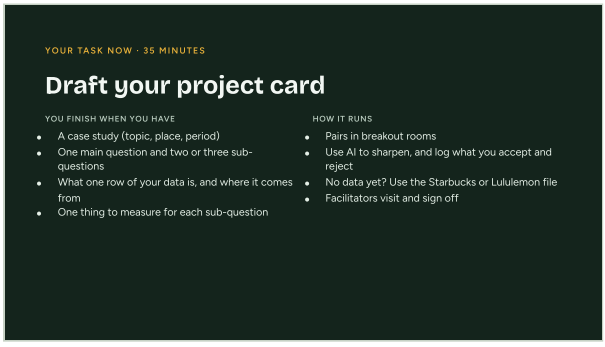
Slide 11

TIME 5 min (0:57 to 1:02)

SAY Step 5: use AI to sharpen your question, not to decide it. Paste in your question and ask it to critique, not to write.

DO Show the prompt on screen and put it in the chat so people can copy it. Remind them: do not paste anything private. Then say: after it answers, write down what you accepted and what you rejected. That is your first verification log entry.

WATCH FOR Students who let the AI rewrite their question completely. Say: it is your question. Keep your words.



Slide 12

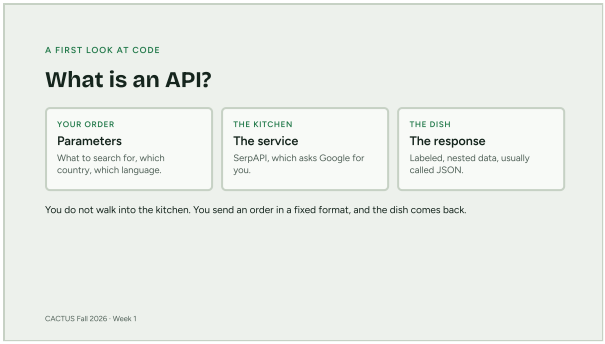
TIME 3 min to introduce (1:02), then the activity (1:05 to 1:40), then 10 min to close (1:40 to 1:50)

SAY Your task now: draft your project card. You finish when you have the four things on the right. 35 minutes, in pairs, in breakout rooms. Facilitators will visit.

DO Put people in breakout rooms of two (or three). Share the card template link in the chat. Tell them the time and when you will call them back. Post a time check at 15 and 30 minutes.

SIGN-OFF (facilitators) Check each card for a case study, a main question with two or three sub-questions, what one row is plus a data source, and one thing to measure per sub-question. Anyone with no data picks the Starbucks or Lululemon file.

AT 1:40 Bring everyone back. Ask two or three volunteers to read their card. Thank them.



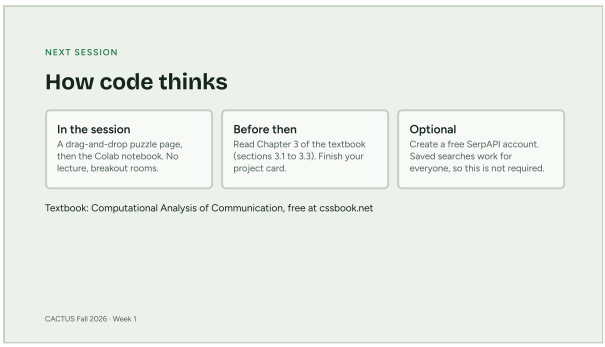
Slide 13

TIME 8 min (1:50 to 1:58)

SAY One last thing before next time: what an API is. Think of ordering at a restaurant. You do not walk into the kitchen. You send an order in a fixed format and the dish comes back.

DO Walk the three cards left to right: parameters are your order, the service is the kitchen, the response is the dish. No code today. Say: next session you will see this with real data.

IF ASKED "What is JSON?" Nested labeled data. You will practice reading it in the puzzle page.



Slide 14

TIME 2 min (1:58 to 2:00)

SAY Next session is about how code thinks. Before then, read Chapter 3 sections 3.1 to 3.3 of the textbook, and finish your project card if you did not today. Creating a SerpAPI account is optional.

DO Post the course site link again. Remind everyone to bring a laptop. Ask if anyone needs help with Google or Colab access, and invite them to message a facilitator privately.

CLOSE Thank everyone. Facilitators stay for five minutes in case anyone has a question.
